

Original Article

Sablefish mortality associated with whale depredation in Alaska

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Killer whale (*Orcinus orca*) and sperm whale (*Physeter macrocephalus*) depredation (whales removing or damaging fish caught on fishing gear) can reduce catch rates and decrease the accuracy of fish stock assessments. This study advances our understanding of the impact of whale depredation on the commercial sablefish (*Anoplopoma fimbria*) fishery in Alaska and evaluates the impact depredation may have on the annual federal sablefish assessment. A statistical modelling approach was used to estimate the whale effect on commercial sablefish fishery catch rates; killer whale depredation was more severe (catch rates declined by 45%–70%) than sperm whale depredation (24%–29%). Total estimated sablefish catch removals 1995–2014 ranged from 1251 t to 2407 t by killer whales in western Alaska management areas and 482 t to 1040 t by sperm whales in the Gulf of Alaska 2001–2014. Including sablefish mortality due to whale depredation on the commercial fishery in the sablefish stock assessment resulted in a 1% reduction in the recommended quota. Accounting for sablefish mortality due to whale depredation in the commercial fishery in the sablefish assessment will occur tandem with correcting for depredation on the annual National Marine Fisheries Service longline survey, the primary survey index used in the assessment.

Keywords: depredation, generalized additive models, killer whales, sablefish, sperm whales.

Introduction

The impact of killer whale (*Orcinus orca*) and sperm whale (*Physeter macrocephalus*) depredation (whales removing or damaging catch caught on fishing gear) on longline commercial fisheries has been evaluated in recent years in two primary regions: (i) the Crozet/Kerguelen Islands and South Georgia Island in the Southern Ocean and (ii) Alaska (Ashford *et al.*, 1996; Roche *et al.*, 2007; Sigler *et al.*, 2008; Clark and Agnew, 2010; Tixier *et al.*, 2010, 2014; Hamer *et al.*, 2012; Peterson *et al.*, 2013, 2014; Straley *et al.*, 2014; Sjöfker *et al.*, 2015). Despite recent advances in our understanding of whale depredation, whale depredation is often not accounted for in fishery stock assessments and the resultant quota or catch share allocation process. This study aims to advance our understanding of the impact of killer whale and sperm whale depredation on the commercial sablefish (*Anoplopoma fimbria*) fishery in Alaskan waters and to evaluate

the impact that including sablefish removals due to whales would have on the annual federal sablefish stock assessment.

The commercial fixed-gear sablefish fishery, which utilizes both pot and hook fishing gear (depredation occurs only on hook fishing gear), operates in Alaskan federal waters from mid-March to mid-November each year; fishing effort tends to be concentrated April to June. Catches (1978–current) in Alaska federal waters ranged from a low of 11 900 metric tons (t) in 2010 to a peak of 38 400 t in 1988. Since 1995 the sablefish directed fishery has been prosecuted as a catch share fishery, with Individual Fishing Quotas (IFQ) issued to quota holders. The sablefish fishery is valuable; the ex-vessel price was U.S. \$2.71/lb in 2013, and in 2011 the sablefish IFQ programme revenue peaked with an ex-vessel value of U.S. \$117 million (Fissel *et al.*, 2013).

The sablefish stock in Alaska is assessed annually by fitting an age-structured model to longline and trawl survey data and com-

mercial sablefish fishery catch data. Standard outputs of the stock assessment include historic and current abundance, fishing mortality, survey selectivity, and catchability (Hanselman *et al.*, 2014). A recommended sablefish catch limit for the following year is calculated by applying a target fishing mortality rate to the estimate of present abundance projected ahead to the next year; the catch limit is then used to determine IFQ limits for the commercial fishery (Hanselman *et al.*, 2014).

Longline survey and commercial fishery sets depredated by killer whales are currently excluded from the sablefish abundance indices used in the stock assessment because catch losses can be severe and highly variable. Conversely, sets impacted by sperm whale depredation are included in the indices because the sperm whale impact on catch rates tends to be less severe (Hanselman *et al.*, 2014). Whale depredation could be incorporated in the sablefish stock assessment model in two primary ways: (i) additional sablefish mortality (whales removing an estimated quantity of fish as part of the commercial fishery) and (ii) expanded survey abundance estimates (more fish would have been caught had whales not depredated).

Killer whale depredation on longline fisheries has a longstanding history in Alaska, with reports from Japanese longliners describing whale interactions in the Bering Sea (BS) dating back to the 1950s (Dalheim, 1988). Killer whale depredation generally occurs in western Alaska: the BS, Aleutian Islands (AI), and Western Gulf of Alaska (WGOA). The issue of killer whale depredation is particularly severe in this region as relatively high densities of fish-eating or resident killer whales overlap with significant longline fishing effort (Fearnbach *et al.*, 2013; Peterson and Carothers, 2013; Peterson *et al.*, 2014). There are limited data on the abundance and population demographics of killer whales in western Alaska; the most recent stock assessment (2013) estimated that a minimum of 1475 resident killer whales use western Alaskan waters; however, these estimates are conservative as they are based on direct counts of animals (Allen and Angliss, 2013a). While it is unknown what proportion of western Alaska resident killer whales depredate on longline fishing gear, statistical modeling approaches comparing catch per unit effort (CPUE) between commercial fishery sets with and without killer whales present have shown that depredating killer whales can reduce sablefish CPUE by as much as 55–70% (Peterson *et al.*, 2013, 2014).

Sperm whale interactions with longline fisheries have been documented since the mid-1980s; however, the frequency of sperm whale depredation events may have increased in the late 1990s following the extension of the sablefish and Pacific halibut (*Hippoglossus stenolepis*) longline fishing seasons associated with IFQ program implementation in 1995 (Hill *et al.*, 1999; Thode *et al.*, 2006; Sigler *et al.*, 2008; Straley *et al.*, 2014). Sperm whale depredation on longline fisheries in Alaska generally occurs in the central and eastern Gulf of Alaska (GOA), although rare cases have been documented as far west as the AI in the commercial fishery and federal longline survey (Hanselman *et al.*, in review). There is no current abundance estimate for sperm whales in north Pacific or Alaskan waters (Allen and Angliss, 2013b). Due to a history of extensive whaling exploitation through the 1970s, sperm whales are currently listed as “endangered” under the 1973 Endangered Species Act and “depleted” under the 1972 Marine Mammal Protection Act.

Depredation by sperm whales in Alaskan waters is presumed to be primarily by solitary mature males from the North Pacific stock (Mesnick *et al.*, 2011). Sperm whale depredation generally has a more moderate effect on fishery or survey catch rates than

killer whale depredation (Peterson and Carothers, 2013). While there are no current estimates for the impact of sperm whale depredation on the commercial sablefish fishery in Alaska, a recent study found that sperm whales depredating on a survey station may reduce the Alaska federal longline survey sablefish CPUE by 12% (Hanselman *et al.*, 2014, in review). In the Southern Ocean, sperm whales can reduce Patagonian toothfish fishery CPUE by as much as 9% (Roche *et al.*, 2007; Tixier *et al.*, 2010).

Whale depredation introduces a negative bias on abundance estimates from the annual NMFS longline survey and estimates of commercial fishery sablefish catches. If CPUE reductions due to whale depredation on the longline survey are to be accounted for in the sablefish stock assessment, it may also be necessary to estimate additional sablefish mortality due to whale depredation on the commercial fishery. Therefore, this study builds upon advances in our understanding of the whale effect on the longline survey (Hanselman *et al.*, in review) to (i) estimate the whale depredation effect on the commercial sablefish fishery CPUE in Alaska, (ii) examine fishery characteristics that may influence the severity of whale depredation, (iii) estimate sablefish mortality attributable to whale depredation, and (iv) estimate the effect of including this mortality as additional fishery landings in the sablefish stock assessment model.

Material and methods

A three-step approach was used to estimate commercial sablefish fishery catch removals associated with whale depredation in Alaska and to evaluate the effects of depredation on the federal sablefish stock assessment and subsequent quota allocations. First, the effect of whale depredation on sablefish CPUE was estimated. Second, the proportion of sets impacted by killer whales and sperm whales was modeled as a function of fishery characteristics to estimate overall catch removals due to whales. And third, the estimated catch removals due to whales were input into the federal sablefish stock assessment.

Estimating the whale effect on sablefish CPUE

The effect of whale depredation on the sablefish commercial fishery CPUE was estimated using National Marine Fisheries Service (NMFS) observer data from 1995 to 2014. From 1995 to 2012, observers were required to monitor approximately one-third of fishing operations of the Alaskan longline fleet in Alaskan federal waters for vessels over 60 ft. in length and to monitor all fishing operations for longline vessels over 125 ft. In 2013 and 2014, the observer programme was restructured such that longline vessels 40 to <60 ft in length were added to the partial observer coverage strata (Alaska Fisheries Science Center, 2015). The basic unit of gear for the NMFS observer data analysis was a set. Each set consisted of one string of hooks ($\bar{x}$ =4817 hooks per set) fished end to end by an observed longline vessel. NMFS onboard observers randomly monitored and recorded species-specific catch data, fishing location information, whale interactions, and general gear performance for each set. Sablefish CPUE was estimated per set as the amount of sablefish caught (kg) per hook.

Each longline set was assigned a performance code (e.g. “no problem”, “considerable whale depredation”, “gear entanglement”, “crab pot in set”, etc.). Only sets (i) with sablefish CPUE greater than zero, and (ii) labelled as “no problem” or “considerable sperm or killer whale predation” as performance codes (32 035 sets) were included in the analysis. Per instructions in the

NMFS observer manual, observers noted if there was “considerable whale depredation” based on visual evidence of whales interacting with the gear and feeding on catch and/or a sudden drop in catch rates (NMFS, 2015). Additionally, observers generally completed a “Comments” section describing the interaction and/or whale species involved. In order to evaluate the use of whale presence as a proxy for depredation, the “Comments” section was manually reviewed for sets labelled as “No Problem” where sperm whales or killer whales were noted as present. Observers also recorded the percentage of a set monitored for marine mammals, as there were instances when the observers were below deck and could not visually monitor for the presence of marine mammals around the fishing gear.

The whale depredation effect on sablefish CPUE was estimated using a generalized additive mixed modelling (GAMM) approach. GAMMs are an extension of linear regression that (i) allow for non-linear relationships between a response variable and explanatory variables based on non-parametric smoothing functions (Wood, 2006; Zuur *et al.*, 2009), and (ii) include fixed and random effects. Whereas fixed effects address the specific effects of a variable at each level, random effects are often associated with batched or grouped data and evaluate the standard deviations or the variations between the different levels (Hanselman *et al.*, in review; Bolker *et al.*, 2009; Zuur *et al.*, 2009).

Sablefish CPUE data were highly skewed and were log-transformed before being modeled as a function of a number of potential covariates (Table 1) in each management area. The final model [Equation (1)] was selected via stepwise regression methods and lowest Akaike Information Criterion (AIC) score (Hardin and Hilbe, 2007; Zuur *et al.*, 2009; Wood, 2015). The coefficient *dep* denotes the effect of depredation, and *D* is an indicator or dummy variable for depredation (*D* = 1 when a set is depredated and *D* = 0 denotes no depredation; Hanselman *et al.*, in review; Peterson *et al.*, 2013) [Equation (1)]. Each model was also fit using the manual review for whale presence (Table 1; coefficient *pres*; *D* = 1 when a set was depredated or had whales present). The maximum degrees of freedom for the smooth terms was restricted to four (with the exception of *lat*, *long*) to accommodate biologically reasonable relationships such as linear, dome-shaped, or asymptotic forms (Goetz *et al.*, 2011; Peterson *et al.*, 2013). All analyses were conducted using R Statistical Software (R Development Core Team, 2014, version 3.1.3), using the “mgcv” package. Random effects were treated as penalized regression terms and were assumed to follow a normal distribution (Wood, 2015).

$$\log(\text{sable CPUE})_{ijk} = f_1(\text{depth}) + f_2(\text{lat}, \text{long}) + f_3(\text{jul}) \\ + f_4(\text{gren_cpue}) + f_5(\text{hal_cpue}) + \text{dep}_i \times D \\ + \text{year}_j + \text{vessel}_k + \varepsilon_{ijk}$$

where the f_i are non-parametric smooth functions and the categorical terms accommodate different average catch rates for year *j* and vessel *k*.

Estimating sablefish catch removals due to whale interactions

We developed a model to predict the number of sets impacted by killer whales or sperm whales by management area. The response variable (number of sets impacted by depredation) was treated as a count variable. NMFS fishery observer data were gridded into

1/3° latitude by 1/3° longitude (~36 km × 25 km; Figure 1) to preserve confidentiality, and a number of fishery attributes was estimated per grid, including average set depth, average number of hooks per set (as a proxy for set length), average sablefish CPUE on non-depredated sets, percentage of haul monitored, average vessel length, and total catch per grid (t).

Estimated sablefish catches (t) per 1/3° grid were based off a catch-in-area trends database that blends processor-based data, mandatory state of Alaska reported landings data, observer data when available, and Vessel Monitoring System data (available 2003–2014). Due to the limited nature of the observer data (partial coverage in many fisheries), these blended data sets are integrated into the NMFS Catch Accounting System to track groundfish fishery harvests annually (Cahalan *et al.*, 2014). Additionally, each grid in the BS, AI, and WGOA was assigned to one of the three “killer whale clusters” per long-term association patterns and geographic ranges, as defined in Fearnbach *et al.* (2013). The three primary clusters evaluated in this analysis, composed of stable matrilineal groups with unique ranging patterns, were based in the western Aleutians (cluster 4), central Aleutians with north/south movements along the BS shelf edge (cluster 2), and the eastern Aleutians and WGOA (cluster 3) (Fearnbach *et al.*, 2013).

Poisson and negative binomial (NB) distributions within generalized linear model (GLM) and generalized additive model (GAM) frameworks were considered to model the response variable, the number of sets depredated per grid. Poisson and NB models are commonly used to model count data; NB models are able to address overdispersion (when the variance is greater than the mean) by adding an additional parameter to model the higher variance (Zeileis *et al.*, 2008; Zuur *et al.*, 2009; Wood, 2015). Zero-inflated poisson (ZIP) and zero-inflated negative binomial GLMs and GAMs were also evaluated due to the high number of zero values in many grids (Zeileis *et al.*, 2008; Zuur *et al.*, 2009; Wood, 2015). The ZIP and NB GAMs consistently fit better than the standard poisson distribution GAMs and GLMs, suggesting evidence of overdispersion and/or zero-inflation in the depredation data. The ZIP GAMs explained the highest percentage of deviance in both management areas and addressed the issues of zero-inflation and false zeros, which was likely a problem with the observer data (not all sets are 100% monitored for marine mammals). Thus, results presented through the remainder of the manuscript will be based on the ZIP GAM models.

The number of sets impacted (*dep_sets*) was modeled as a function of potential explanatory variables averaged or summed per grid. The total number of sets (*total_sets*) per grid was included as an offset to account for variations in fishing effort per grid. When significant collinearity occurred (Pearson’s correlation test; $r > 0.5$, $p < 0.05$), one of the two variables was dropped from the final model based on lowest AIC score (Bonate, 1999).

The final model [Equation (2)] for the western region (Western areas BS, AI, and WGOA), 1995–2014, $n = 2801$ grid/year combinations, Table 2) included the following terms:

$$\text{dep_sets}_i = f_1(\text{avg_cpue}) + f_2(\text{catch}) + f_3(\text{avg_length}) \\ + f_4(\text{avg_depth}) + f_5(\text{lat}, \text{long}) + \text{year}_i \\ + \log(\text{total_sets}) + \varepsilon_i. \quad (2)$$

There were no observations of sperm whale depredation on the commercial fishery in the Central Gulf (CG), West Yakutat

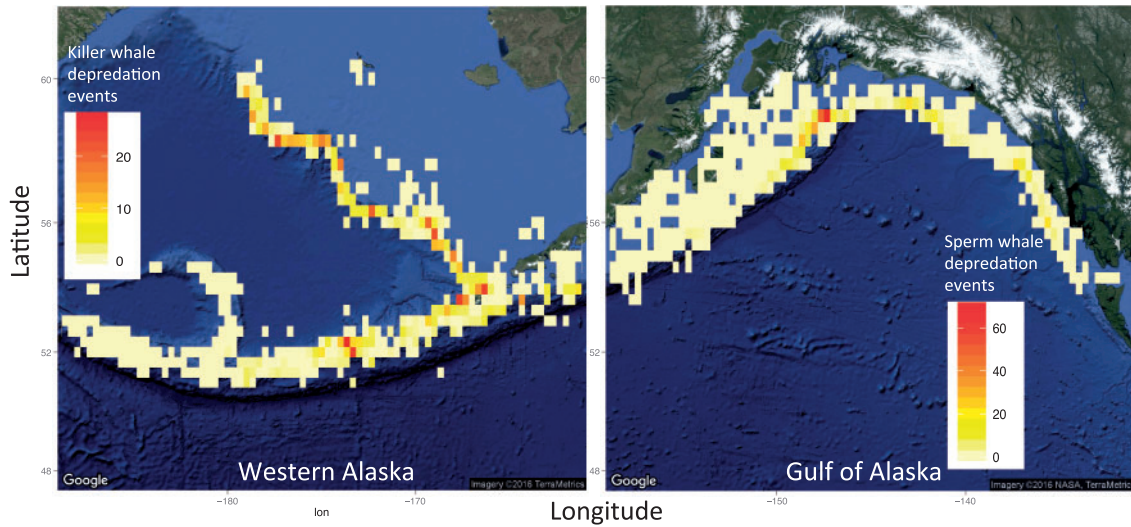


Figure 1. Depredation event counts in $1/3^\circ$ by $1/3^\circ$ grid cells by killer whales in western Alaska and sperm whales in the GOA.

(WY), and Southeast (SE) prior to 2001, so only years 2001–2014 were included in the eastern management area (East) analysis. The final model [Equation (3)] for the East region ($n=1600$ grid/year combinations, Table 2) included the following terms:

$$\text{dep_sets}_i = f_1(\text{avg_length}) + f_2(\text{lat}, \text{long}) + \text{year}_i + \log(\text{total_sets}) + \varepsilon_i. \quad (3)$$

The total weight of sablefish taken by depredating whales was estimated based on the proportion of sets depredated and average CPUE reductions on depredated sets. The model-predicted number of sets impacted by whales including year effects was applied as a proportion of the total sets ($\widehat{\text{prop_dep}}$) to the estimated sablefish landings (catch_i) and model-estimated CPUE reductions by area ($\widehat{\text{prop_reduct}}$, as estimated in the section “Estimating the whale effect on sablefish CPUE”, Table 2). These terms were multiplied to estimate additional sablefish mortality due to whale depredation on the commercial sablefish fishery.

$$\text{sable_mort} = \widehat{\text{prop_dep}} * \text{catch}_i * \left(\frac{1}{(1 - \widehat{\text{prop_reduct}})} - 1 \right). \quad (4)$$

Average area model-estimates for CPUE reductions and proportion of sets impacted by depredation (Tables 3 and 4) were extended to grids with extant catch data (but no observer or depredation data) for years when depredation occurred. This approach necessitated the assumption that certain fishery characteristics, such as vessel size or amount of gear set, stayed relatively constant across these unobserved grids.

Fishery characteristics affecting depredation

In order to evaluate spatial and fishery characteristics that affect depredation, the commercial fishery data were further aggregated across years. Equations (2) (West) and (3) (East) were modified to fit the data without a year effect using the ZIP GAM (Wood, 2015). The final models with the aggregated data in the West

($n=421$ grids) and East ($n=342$ grids) 1995–2014 took the following forms:

$$\begin{aligned} \text{dep_sets}_i = & f_1(\text{avg_cpue}) + f_2(\text{catch}) + f_3(\text{avg_hooks}) \\ & + f_4(\text{lat}, \text{long}) + \text{cluster}_i + \log(\text{total_sets}) + \varepsilon_i \end{aligned} \quad (5, \text{West})$$

$$\begin{aligned} \text{dep_sets}_i = & f_1(\text{catch}) + f_2(\text{avg_length}) + f_3(\text{lat}, \text{long}) \\ & + \log(\text{total_sets}) + \varepsilon_i \end{aligned} \quad (6, \text{East})$$

Sablefish stock assessment application

The Alaska federal sablefish assessment model is a one-stock model that assumes the BS, AI, and the GOA are well mixed based on an extensive mark-recapture data set indicating high movement (Hanselman *et al.*, 2014). The stock is assessed annually by fitting an age-structured model to survey and commercial fishery catch rates (which are treated as relative indices of abundance) and to commercial fishery and survey catch age and length compositions (Hanselman *et al.*, 2014). Catch data are treated as lognormal; age, and length compositional data are treated as multinomial. Natural mortality and growth schedule (mean and variance of length at age) are estimated externally. The model fit includes estimates of present and historical abundance (including annual recruitments), fishing mortalities, and commercial fishery and survey selectivities and catchabilities. The recommended catch limit is estimated by projecting the numbers at age one year forward to set the Acceptable Biological Catch (ABC), which in turn informs the IFQ limits for the commercial fishery.

To date, the additional sablefish removals associated with whale depredation during the survey and/or the commercial fishery have not been accounted for in the stock assessment model; therefore, there is likely some bias in model outputs (because survey catch rates may seem artificially low, and because total catch/mortality is underestimated). We investigated the effects on the sablefish stock assessment of accounting for whale depredation in the fishery by fitting the model with and without the extra sablefish mortality due to whales (based on catch removal estimates

Table 1. Explanatory variables considered, whether they were included as continuous variables, smoother terms, factors or fixed or random terms, and their description.

Explanatory terms	Model input	Significant ($p < 0.05$)	Descriptor
Year	Factor	Y	1995–2014, March to November
Vessel	Factor	Y	Unique vessel identification
Depth (meters)	Smoother	Y	Estimated bottom depth (meters)
Julian day	Smoother	Y	Continuous count of days in a given year
Vessel length (feet)	Smoother	N	Length of vessel fishing (feet)
Grenadier CPUE (kg/hook)	Smoother	Y	Grenadier catch per unit effort (kg/hook)
Pacific halibut CPUE (kg/hook)	Smoother	Y	Pacific halibut catch per unit effort (kg/hook)
Whale depredation (Yes/No)	Factor	Y	Presence of depredating killer whales (BS, AI, WGOA) or sperm whales (CG, WY, SE) with observer noting evidence of whales feeding on catch (evidence)
Whale presence (Yes/No)	Factor	Y	Sets tagged as depredated by onboard observers + sets with sperm whales or killer whales present based on visual sighting and manual review of observer comments (evidence and/or presence)
Latitude, Longitude	Smoother	Y	Coordinates for each set sampled

The response variable was log-transformed sablefish CPUE (kg/hook) for sets with sablefish CPUE > 0, 32 034 sets fished, 1995–2014.

Table 2. Explanatory variables considered, whether they were included as smoother terms or factors, their significance in western Alaska (West) and eastern Alaska (East) models, and their description.

Explanatory terms	Model input	Significant western Alaska (West; $p < 0.05$)	Significant eastern Alaska (East; $p < 0.05$)	Descriptor
Location (<i>lat, long</i>)	Smoother	Y	Y	Latitude and longitude rounded to 1/3° by 1/3° to create grids
Year (<i>year</i>)	Factor	Y	Y	1995–2014 for western Alaska (killer whale depredation); 2001–2014 for eastern Alaska (sperm whale depredation)
Set depth (<i>avg_depth</i>)	Smoother	Y	N	Estimated average bottom depth per grid (m)
Number of hooks per set (<i>avg_hooks</i>)	Smoother	Y	N	Average number of hooks per set per grid grid
Sablefish CPUE (<i>avg_cpue</i>)	Smoother	Y	N	Average sablefish CPUE (kg/hook) per grid on non-depredated sets only
Total catch (<i>catch</i>)	Smoother	Y	N	Total sablefish catch per grid (t)
Vessel length (<i>avg_length</i>)	Smoother	Y	Y	Average length of vessels fishing (feet) per grid
Percentage haul monitored (<i>%haul_mon</i>)	Smoother	N	N	Average percentage of haul monitored for whales by onboard fisheries observer per grid
Whale cluster (<i>cluster</i>)	Factor	N	NA	Killer whale cluster as defined by Fearnbach et al. (2013) for the BS, AI, WG grids only

The response variable was the total number of depredated sets in a 1/3° by 1/3° grid, West: 2801 grid/year combinations 1995–2014, East: 1 600 grid/year combinations 2001–2014.

from the section “Commercial fishery catch removals due to whale interactions”). Years 1995–2001 were not included in the East model due to limited observed depredation events pre-2002. For 1995–2001 and 2015, we used the average sperm whale depredation estimate to fill in those missing years. The additional estimated mortality from both species of whales in all areas was combined with the longline fishery catch in the model (where depredation occurs), which assumes that whales take a representative sample of the longline catch (i.e. they do not selectively target large fish).

Results

The whale effect on sablefish CPUE

Depredating killer whales in the BS, AI, and WGOA management areas caused the greatest commercial fishery CPUE reductions: 45.7–69.4% (Table 3). Sperm whale-associated CPUE reductions

were relatively less severe, ranging from 23.8 to 29.4% in the CG, WY, and SE management areas (Table 3). Estimates of CPUE reductions associated with the manual review process tended to be lower than the standard estimates (~1–6% lower), with the exception of the AI management region (Table 3).

Commercial fishery catch removals due to whale interactions

For gridded cells with observer information available, the observed percentage of sets depredated ranged from 3.9% in the WGOA to a high of 16.4% in the BS. The proportion of sets depredated by killer whales in the West varied between years, but did not increase or decrease significantly between years (linear regression; $p > 0.05$; Figure 2a). The model-estimated percentage of sets impacted was generally slightly lower than the observed values in

Table 3. GAMM results by management area and whale depredation term.

Area	Depredation term	Depredation coefficient (% CPUE red)	Confidence interval ($\pm 2^*SE$)	DF	n	AIC	%dev
Bering Sea	KW	45.7%	34.7–56.6%	103	4 339	13 064	49.7%
	KW man rev	44.6%	34.1–55.2%			13 063	49.8%
Aleutians	KW	57.7%	42.6–72.7%	101	6 744	17 612	37.2%
	KW man rev	59.6%	45.3–73.9%			17 583	37.5%
WGOA	KW	69.4%	56.5–82.1%	103	5 950	13 028	31.0%
	KW man rev	68.3%	55.6–80.7%			13 023	31.1%
Central GOA	SW	23.8%	15.1–32.4%	193	8 218	17 281	46.4%
	SW man rev	19.9%	12.8–27.0%			17 281	46.4%
West Yakutat	SW	26.3%	16.6–36.0%	119	3 919	6 525	52.7%
	SW man rev	18.0%	10.8–25.2%			6 534	52.6%
Southeast	SW	29.4%	15.8–43.0%	124	2 865	5 692	43.5%
	SW man rev	23.2%	12.6–33.7%			5 693	43.5%

The response variable, CPUE (kg/hook) for sets with sablefish CPUE > 0, followed normal distribution. The results display the depredation coefficient or the model-estimated difference in catch between depredated and non-depredated sets, with 95% CI as 2^*SE , degrees of freedom (DF), the sample size for a given area (n), Akaike Information Criterion (AIC), percentage of deviance explained (%dev).

KW, killer whale depredation; KW man rev, manual review of observer comments; SW, sperm whale depredation; SW man review, manual review of observer comments.

the West: $12.6 \pm 5.5\%$ (2^* Standard Error) in the BS, $4.2 \pm 1.9\%$ in the WGOA, and $4.1 \pm 2.0\%$ in the AI (Table 4).

Total estimated catch removals during 1995–2014 due to killer whales ranged from 1251.6 ± 371.0 t in the AI to a high of 2407.5 ± 400.6 t in the WGOA (Table 4). If sablefish removals due to killer whales were added to the overall catch reported in the NMFS Catch Accounting System, the total catch removals due to would increase by 7.1% in the AI, 12.1% in the BS, and 7.9% in the WGOA (Table 4). The overall sablefish catch removals (t) due to killer whales did not increase significantly over time (linear regression; $p > 0.05$; Figure 3).

The observed proportion of sets depredated by sperm whales in the East increased significantly from 2001 to 2014 (linear regression; $p < 0.05$; Figure 2b). The predicted percentage of sets impacted was relatively low in all three East management areas from 2001 to 2014 ($2.7 \pm 1.5\%$ in the CG, $4.7 \pm 2.5\%$ WY, $4.4 \pm 2.6\%$ SE; Table 4). Despite the low percentage of sets impacted, catch removals due to sperm whales were still moderate due to higher overall catches in the East historically: 1040.3 ± 400.6 t in the CG, 482.3 ± 189.4 t in WY, 822.2 ± 352.1 t in SE (Table 4 and Figure 3). Whale-associated catch removals were minimal in comparison to overall fishery catches in the East ($\sim 1\%$; Table 4) and increased moderately from 2001 to 2014 ($p = 0.51$).

Fishery characteristics affecting depredation

Killer whales targeted areas in the West with higher average sablefish CPUE (up to about 0.08 kg/hook), areas of greater total sablefish catches, and longer sets with more hooks on average (Figure 4). Each of these relationships took a dome-shaped form (Figure 4), suggesting limitations of the effect these covariates have on the number of sets depredated. Killer whale depredation was most severe along the slope edge in the BS between -171° and -176° longitude (Figure 4). Cluster 3 whales (eastern Aleutians and WGOA) were more likely to depredate than Cluster 2 or Cluster 4 whales (Figure 4). The model fit the data well (%Deviance explained = 84.4). Sperm whales in the East targeted areas with higher catches, areas with longer vessels on average, and the zone between -145° and -150° longitude (Figure 5).

The model also fit the data well in the East (%Deviance explained = 93.9).

Sablefish stock assessment application

Estimated catch removals due to depredating whales varied by year and did not show a significant increasing or decreasing trend over the time period (Figure 6). These estimates of additional mortality due to whale depredation were included as additional catch in the fixed gear fishery in two comparisons with the assessment model used for management. While the effects of whales were substantial at localized events, the total number of these events was small.

The incorporation of these corrected catch estimates in the stock assessment models resulted in a very small divergence in estimates of female spawning biomass (SSB) from the uncorrected reference model (Figure 7). The maximum estimated difference between the reference model was slightly greater than 1% and peaked in 1996 and generally declined until the end of the time series. Other key model results were very similar; sablefish stock productivity increased moderately and resulted in a slightly higher average recruitment (Figure 8).

Discussion

The whale effect on sablefish CPUE

Consistent with other studies from Alaska, depredating killer whales have a significant effect on the commercial sablefish fishery CPUE (45.7–69.4% reductions, Table 3). In previous studies, killer whale depredation was estimated to reduce sablefish CPUE by 54–72% on depredated longline survey sets, and 35–69% reductions for the commercial sablefish fishery (Peterson *et al.*, 2013, 2014). Estimates of CPUE reductions due to killer whales in Alaska are somewhat higher than estimates from the Crozet Exclusive Economic Zone (EEZ) in the Southern Ocean, where killer whales can reduce Patagonian toothfish (*Dissostichus eleginoides*) CPUE by as much as $27 \pm 25\%$ (Roche *et al.*, 2007; Tixier *et al.*, 2010). It may be that killer whale depredation on longline fisheries in western Alaska has a more longstanding history than in the Crozet EEZ (where depredation began in the late 1990s) and that depredating killer whales in Alaska are fish-eating

Table 4. Model results, model-estimated proportion of sets impacted (prop dep), associated sablefish catch removals (t), and the proportion increase in overall sablefish removals (prop overall catch) due to killer whales by management area, 1995–2014 and sperm whales by management area, 2001–2014.

Area—Whale	ZIP GAM Model Structure	df	%Dev exp	AI prop				BS prop				WG				WG prop				Western Alaska catch removals			
				AI prop dep	AI catch removals	AI prop overall catch	BS prop dep	BS prop overall catch	WG prop dep	WG catch removals	WG prop overall catch	WG prop dep	WG catch removals	WG prop overall catch	WG prop overall catch	WG prop overall catch	WG prop overall catch	WG prop overall catch	WG prop overall catch	WG prop overall catch	WG prop overall catch	WG prop overall catch	WG prop overall catch
Western Alaska—Killer whale	$dep_{sets_i} - f_1(avg_cpue) + f_2(catch) + f_3(avg_length) + f_4(avg_depth) + f_5(lat, long) + year_i + \log(total_sets) + \epsilon_i$	51	43.7	0.041 ± 0.02	1251.6 ± 371.0	0.071 ± 0.02	0.126 ± 0.055	0.121 ± 0.04	0.042 ± 0.019	2407.5 ± 400.6	0.079 ± 0.01	0.042 ± 0.019	2407.5 ± 400.6	0.079 ± 0.01	0.079 ± 0.01	0.079 ± 0.01	0.079 ± 0.01	0.079 ± 0.01	0.079 ± 0.01	0.079 ± 0.01	0.079 ± 0.01	0.079 ± 0.01	0.079 ± 0.01
Eastern Alaska—Sperm whale	$dep_{sets_i} - f_1(avg_length) + f_2(lat, long) + year_i + \log(total_sets) + \epsilon_i$	35	58.8	0.027 ± 0.015	1040.3 ± 400.6	0.009 ± 0.003	0.044 ± 0.026	0.012 ± 0.005	0.047 ± 0.025	482.3 ± 189.4	0.012 ± 0.005	0.047 ± 0.025	482.3 ± 189.4	0.012 ± 0.005	0.012 ± 0.005	0.012 ± 0.005	0.012 ± 0.005	0.012 ± 0.005	0.012 ± 0.005	0.012 ± 0.005	0.012 ± 0.005	0.012 ± 0.005	0.012 ± 0.005

Depredation and fishery data were divided into 1/3° by 1/3° grids. Model results are presented for the response variable, number of sets depredated, following a zero-inflated Poisson (ZIP) distribution in a Generalized Additive Model (GAM) framework. Results show: degrees of freedom (df), percentage deviance explained (%Dev Exp), the model-estimated proportion of sets depredated (Prop Dep) ± 2 * SE, catch removals (t) ± 2 * SE by management area and whale species, and the proportion increase in overall catches due to whales (prop overall catch) ± 2 * SE.

specialists, whereas depredating killer whales in the Crozet EEZ are generalists, feeding on fish, sea birds, seals and whales (Dalheim, 1988; Matkin and Saulitis, 1994; Yano and Dalheim, 1995; Fearnbach *et al.*, 2013; Tixier *et al.*, 2015).

The greatest CPUE reduction due to killer whales was seen in the WGOA (Table 3) despite the relatively low percentage of sets impacted (~4–6%). The greater reductions in the WGOA may be associated with higher average sablefish CPUEs in the WGOA (two to five times greater than AI or BS sablefish CPUE, respectively, such that killer whales may be targeting areas where there is a better opportunity to take more fish).

Reductions in sablefish CPUE due to sperm whale depredation were more moderate than killer whale depredation (Table 3). Sperm whales may be less efficient at removing fish than killer whales, and sperm whales also tend to depredate as solitary individuals as opposed to killer whales that often depredate in groups (Sigler *et al.*, 2008; Tixier *et al.*, 2010; Fearnbach *et al.*, 2013; Straley *et al.*, 2014; Söffker *et al.* 2015). This study represents the first time that depredation has been evaluated at this level with commercial sablefish fishery data, and the estimates of sperm whale-associated CPUE reductions on the commercial fishery are about twice as high as estimates of the sperm whale effect on the NMFS longline survey (Hanselman *et al.*, in review). This commercial fishery analysis excluded sets with sablefish CPUE equal to zero; thus, the signal of the sperm whale effect may be stronger with low sablefish CPUE sets excluded. The vessels used to conduct the longline survey are large factory boats that process all of the catch, unlike the smaller catcher vessels that participate in the sablefish fishery and often only bleed the catch. The substantial trail of offal generated is commonly fed on by sperm whales and may distract them from targeting the line.

Depredation can be difficult to confirm based on evidence of damaged fish. Observers are often forced to make an arguably subjective call as to whether or not “considerable whale depredation” occurred on a set. In order to address potential inconsistencies with identifying depredated sets, we also conducted a manual review of the comments sections for sets where sperm or killer whales were present, but the sets were not labelled as depredated. These additional sets were labelled as “whales present and potentially depredating (*pres*).” The same models were run with this manual review term (*pres*) that included sets labelled depredated and sets with whales present. The manual review term still showed a significant sperm whale and killer whale effect on CPUE in each area (Table 3); however, the whale effect was slightly lower for killer whales in the BS and WGOA and increased moderately in the AI (Table 3). The change in whale effect on CPUE was greater in the East for sperm whales with the manual review term (Table 3). This highlights that sperm whales may be more likely to be present around a fishing vessel and not necessarily depredating. For instance, sperm whales have been shown to feed on offal or discards as opposed to hooked catch on the gear (Mathias *et al.*, 2012). Conversely, the changes in the killer whale effect on CPUE associated with the manual review term were slight, suggesting that killer whales present are more than likely depredating on fishing gear.

Catch removals due to whale interactions

Neither the proportion of sets impacted by killer whales nor the associated catch removals increased significantly over the study

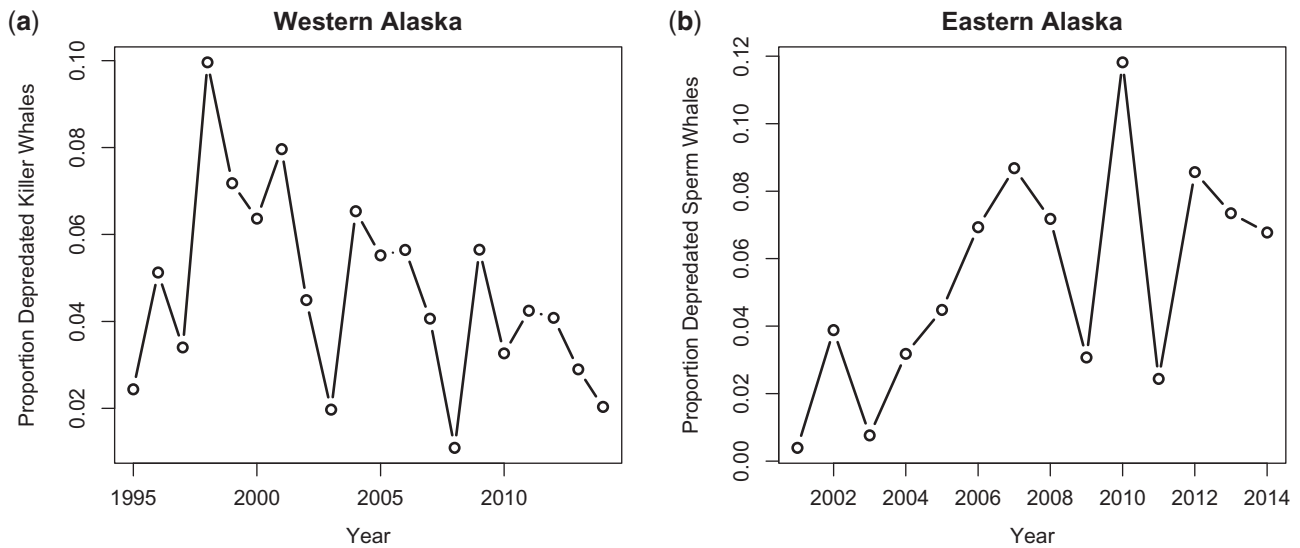


Figure 2. The observed proportion of sets depredated by killer whales in western Alaska, 1995–2014 (a) and by sperm whales in eastern Alaska, 2001–2014 (b).

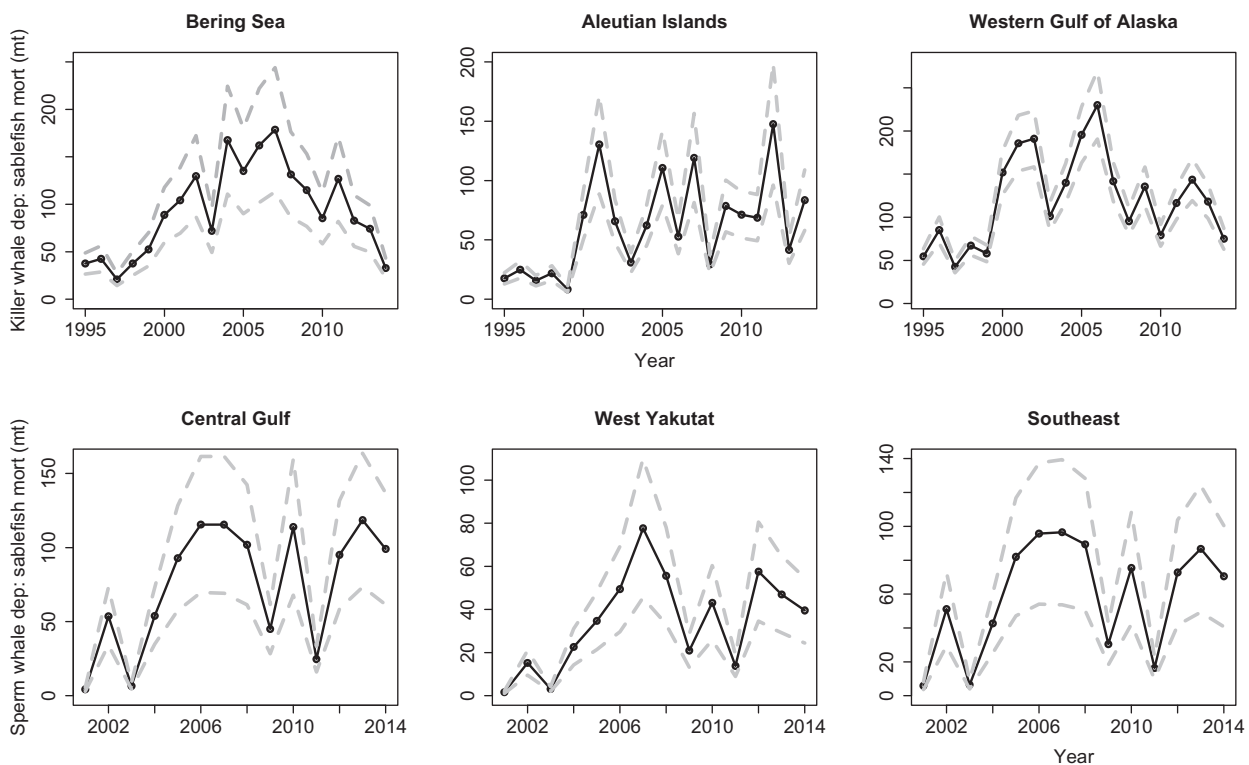


Figure 3. Estimated sablefish mortality (t) by year due to killer whales in the BS, AI, and WGOA and sperm whales in the Central GOA, West Yakutat, and Southeast Alaska with ~95% confidence bands.

period. This is consistent with the longer history of killer whale depredation in western Alaska and suggests that the depredation behavior may not be expanding in this area. The effects of depredation on overall catches did vary by management area, and it is important to note that killer whale depredation may be responsible for removing a significant portion of the overall catch by management area in western Alaska management areas, especially

the BS. The sablefish fishery in the BS is not fully prosecuted in most recent years, however, and if whale-associated removals were to be included in overall fixed gear catches, the fishery in the BS would still be under the area-apportioned quota.

Sperm whale depredation in the East was not documented by onboard observers until 2001. It is likely, however, that sperm whale depredation did occur on the commercial fishery prior to

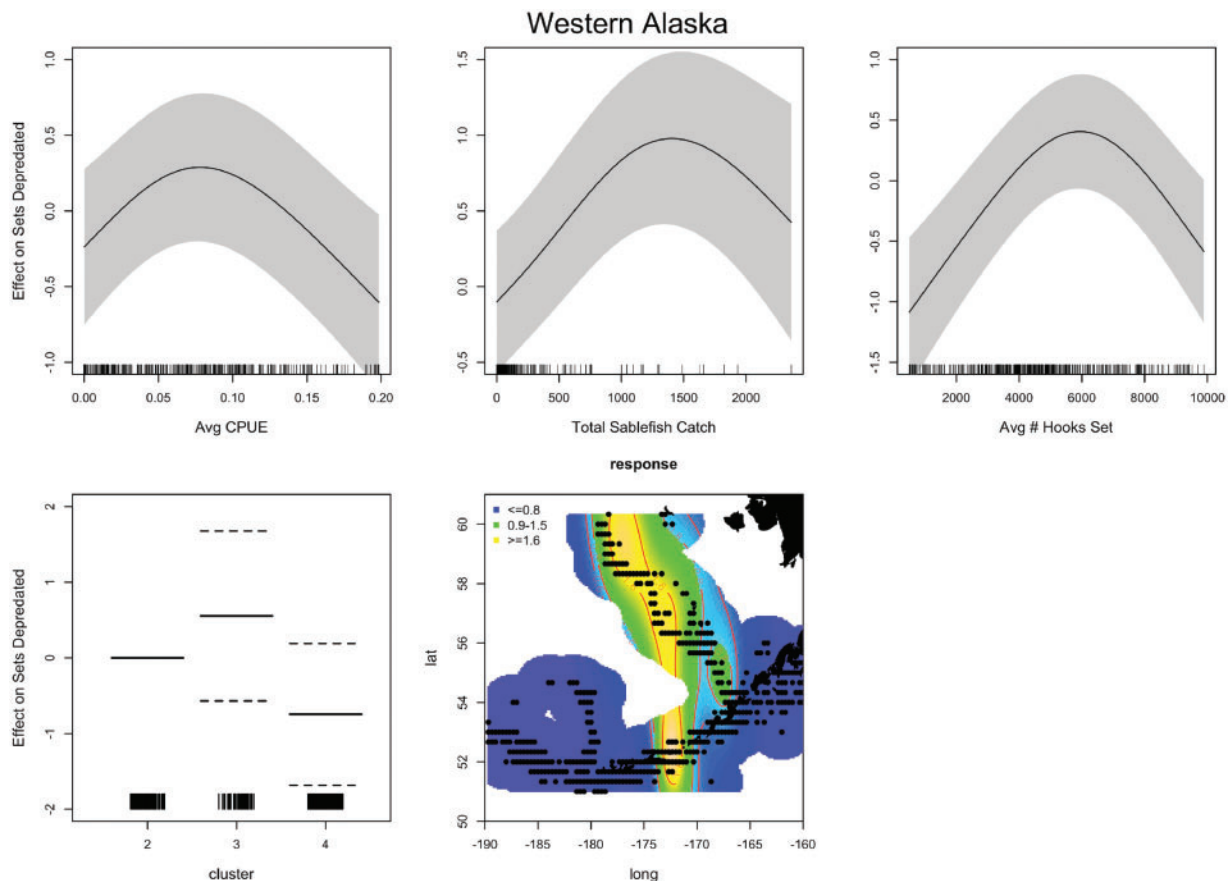


Figure 4. Zero-inflated Poisson GAM model results for western Alaska, $n = 421$, %Dev = 84.4%. Additive effects of average CPUE, total sablefish catch, average number of hooks per set, whale cluster (cluster 4 western Aleutians 4; cluster 2 central Aleutians, with north/south movements along the Bering Sea shelf edge; cluster 3 eastern Aleutians and WGOA), and spatial location (latitude, longitude) on the number of sets depredated by killer whales in the BS, AI, and WGOA. Shaded areas represent $\sim 95\%$ confidence bands.

2001 (but was not identified by onboard observers) as it was documented on the longline survey as early as 1998 (Hanselman *et al.*, in review). Since 2001, catch removals associated with sperm whale depredation increased moderately, and they may be taking a greater proportion of the catch over time because catch limits have declined in recent years. In regions where the area-specific fisheries are fully prosecuted each year (CG, WY, SE), sperm whales taking a relatively greater portion of the overall catch over time could become an issue in light of concerns over recent declines in the sablefish stock biomass.

The observed number of sets impacted by sperm whales and killer whales was likely biased low; therefore, a zero-inflated model was used to address the issue of “false zeros”, where observer error could have resulted in depredated sets not being designated properly (Zuur *et al.*, 2009). A number of factors could account for a low bias in the observed proportion of sets labelled as depredated: (i) the average observer coverage rate for the sablefish fishery was relatively low from 1995 to 2014 (approximately one-third sets monitored on vessels >60 ft or roughly 14% of landed catch), (ii) observers were often required to be sampling below deck and could not confirm the presence of marine mammals, and (iii) fishermen sometimes opt to wait to haul the remainder of their gear until the whales have left the fishing grounds (partially impacted sets not always accounted for). Due

to these inconsistencies associated with identifying depredated sets with limited observer coverage, the uncertainty of these estimated catch removals is likely underestimated.

Fishery characteristics affecting whale depredation

The opportunistic nature of killer whale foraging ecology in other contexts (Herman *et al.*, 2005; Krahn *et al.*, 2007; Matkin *et al.*, 2007) likely plays a major role in determining which fishery characteristics whales may target. The number of depredated sets had a dome-shaped relationship with sablefish cpue, sablefish catch, and average fishing effort, suggesting that killer whales are less likely to target areas with very low sablefish CPUE, but that they are also highly opportunistic foragers that will take advantage of fishing operations when access is available. Additionally, killer whales were targeting areas where more gear is set in the water (sets with more hooks). Larger vessels (>120 ft) on average tended to set longer sets with more hooks (Pearson's correlation test; $r=0.66$, $p < 0.001$). These findings concur with research in the Southern Ocean that found that shorter sets can be an effective method for fishermen to reduce killer whale depredation (Tixier *et al.*, 2014).

Killer whale depredation was most likely to occur along the BS slope edge between -171° and -176° longitude (Figure 3).

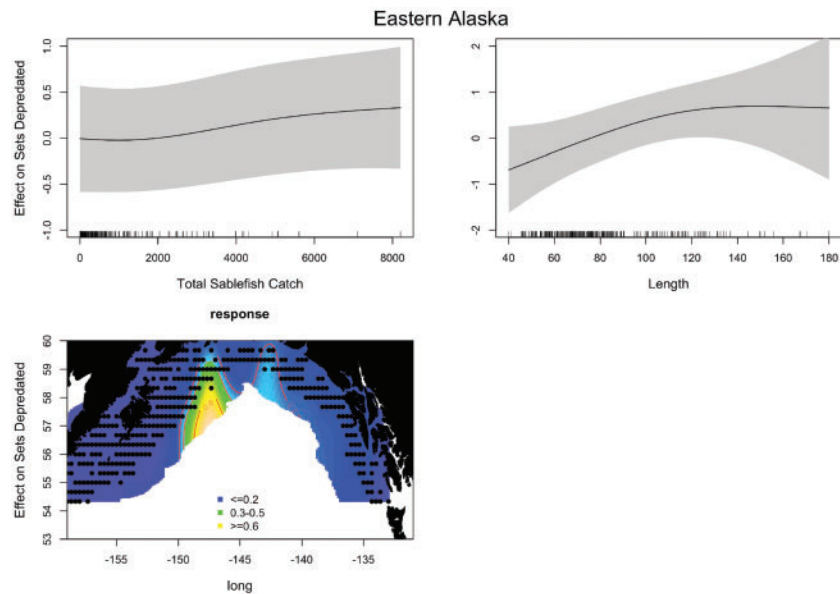


Figure 5. Zero-inflated Poisson GAM model results eastern Alaska, $n = 338$, %Dev = 93.9%. Additive effects of total sablefish catch per grid, average vessel length, and spatial location (latitude, longitude) on the number of sets depredated by sperm whales in the CG, WY, and SE. Shaded areas represent $\sim 95\%$ confidence bands.

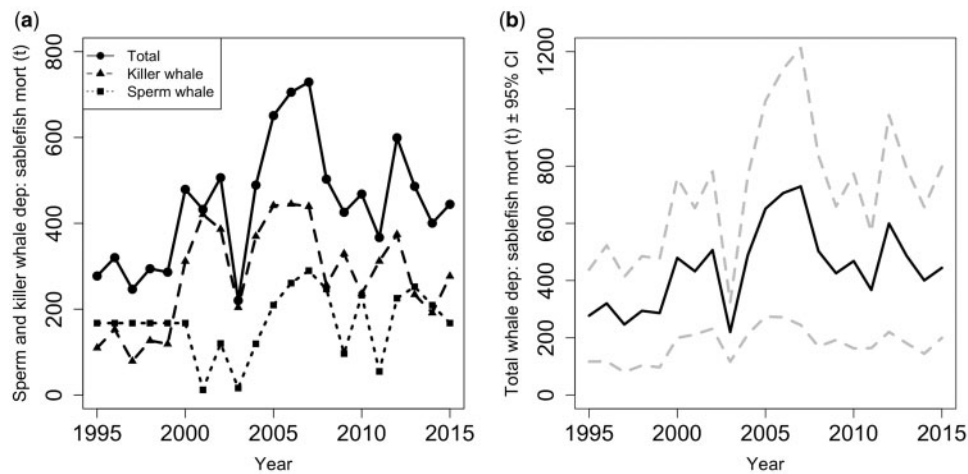


Figure 6. Estimated sablefish mortality (t) due to sperm whale and killer whale depredation 1995–2015. Additional estimated sablefish mortality by whale species (a), and total estimated sablefish mortality due to whale depredation by year with 95% asymptotic normal confidence intervals (b).

These results concur with an earlier study evaluating the likelihood of killer whale depredation on federal longline survey stations along the BS slope (Peterson *et al.*, 2013). The whale cluster term was significant in the western Alaska model, and Cluster 3 whales (eastern Aleutians and WGOA) were more likely to depredate than Cluster 2 or Cluster 4 whales (Figure 4). The significant difference between cluster 3 whales and the other two clusters may also be linked to the greater reductions in CPUE in the WGOA due to killer whale depredation (Table 3). Cluster 3 whales were shown to on average make the largest movements of any cluster (median range = 430 km), and this larger range may allow for easier transmission of the depredation behavior amongst associated matriline (Fearnbach *et al.*, 2013).

Although not significant, the number of sets depredated increased with the percentage of the haul monitored by onboard observers ($p > 0.05$); this logical relationship highlights that whale depredation may be underrepresented in instances when observers are unable to monitor for whales as well as sample fish. For instance, in some years over 50% of sets on the NMFS federal longline survey have been depredated by killer whales in the BS alone, and fishermen's accounts from the region also suggest that the observer data may underrepresent the depredation issue in Alaska (Peterson and Carothers, 2013; Peterson *et al.*, 2013; Hanselman *et al.*, 2014). Thus, it is important that statistical approaches be used to estimate upper bounds on confidence intervals or valid approaches for addressing false zeros to best reflect the full scope of depredation and potential impacts to the

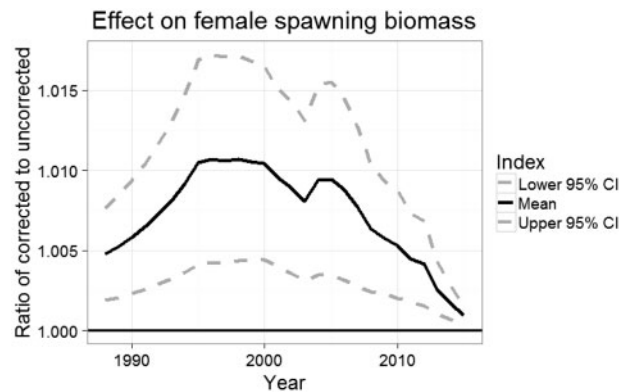


Figure 7. The ratio of female spawning biomass estimates from sablefish assessment models that include additional mortality from whale depredation in the fishery to the assessment model used for management. Mean (solid line) is the point estimate, Dashed lines are models including the lower and upper confidence intervals of the estimated additional sablefish mortality from whale depredation.

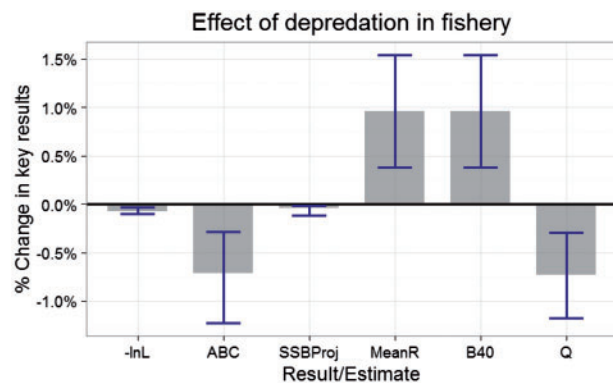


Figure 8. Change in key results in the Alaska sablefish assessment when whale depredation is accounted for in the fishery; “-lnL” is the negative log likelihood, “ABC” is the Acceptable Biological Catch recommendation, “SSBProj” is the female spawning biomass projected for the following year, “MeanR” is the average recruitment, B40 is the estimated $B_{40\%}$ target reference point, and “Q” is the catchability or proportionality constant for the longline survey abundance index. Error bars are models including the lower and upper confidence intervals of the estimated additional sablefish mortality from whale depredation.

fish stock and commercial fishery. This study addressed this by incorporating zero-inflated models. Additionally, the catch could be further expanded to an upper confidence interval of the estimates as a precautionary measure.

Sperm whale depredation was more common in grid areas with more overall sablefish catch (t) and longer vessels on average (ft, Figure 5). Sablefish CPUE was not significant and did not improve model performance in the East. This highlights an interesting potential difference in the behavior of depredating sperm whales. Sperm whales may target zones with more fishing effort (more vessels fishing longer sets) as opposed to more productive sets with higher CPUEs. Additionally, average sablefish CPUE in the East is greater than average CPUE in the West, suggesting there may be a “threshold effect,” beyond which additional

increases in CPUE in the East are not critical for attracting depredating sperm whales because CPUEs are already relatively high. Sperm whales are generally thought to be cueing in on fishing vessel noise underwater, especially the cavitation noise generated by changes in the propeller rotation (Thode *et al.*, 2007). Higher catches suggests more fishing activity in certain grids and more acoustic cues to motivate sperm whale depredation.

Sablefish stock assessment application

Inclusion of the additional mortality due to whales in the stock assessment model resulted in very small differences in assessment estimates of key management quantities (e.g. 1% reduction in recommended quota). Including the unaccounted mortality showed that the stock was moderately more productive than previously thought, resulting in a slightly higher average recruitment and target reference point. While the sablefish assessment model had a very low retrospective trend (Hanselman *et al.*, 2014), the model tended to slightly overestimate spawning biomass each year, possibly indicating some unaccounted for mortality associated with whale depredation.

It remains unresolved when and whether to utilize these corrected catches in the sablefish stock assessment. First, the extent of killer whale depredation prior to 1995 in the fishery is unknown, but is well known to have occurred historically. Conversely, there has been an apparent increase in sperm whale depredation in recent years. Historically depredation likely had a minor impact, but it is an added uncertainty. The effect of sperm whales on the NMFS longline survey and resultant survey abundance index, while smaller at each depredation event, is more widespread and had a greater effect on the stock assessment when accounted for (Hanselman *et al.*, in review). Inclusion of whale depredation in fishery removals had a slightly negative effect (<1%) on the recommended catch limit, while the inclusion of survey depredation had a larger positive effect (~5%, Hanselman *et al.*, in review).

Conclusions and implications

Sperm whale and killer whale depredation has a significant impact on the commercial sablefish fishery through reduced CPUEs, increased (and unaccounted for) sablefish mortality, reduced accuracy for sablefish abundance indices, and increased operating costs for the commercial fishery (Peterson and Carothers, 2013; Peterson *et al.*, 2014). Although there may be some disadvantages for depredating whales, such as increased risk of vessel strikes or entanglement in fishing gear (Allen and Angliss, 2013a), the persistence of whale depredation in Alaska over the last 20 years despite attempts by fishermen to minimize these interactions suggests the benefits outweigh the risks for these adaptable and opportunistic predators. In the Crozet archipelago in the Southern Ocean, “artificial food provisioning” by depredating killer whales had a positive effect on female calving rates (Tixier *et al.*, 2015).

While research into acoustic deterrents and depredation mitigation devices is ongoing (Hamer, 2015; Straley *et al.*, 2015), there are currently no proven effective deterrent options for mitigating depredation on demersal longline fishing operations. Fishing operations and management of the sablefish commercial fishery have adapted in other ways; however, pot fishing gear for sablefish (which can not be depredated by whales) is legal in the AI and BS regions. From 2004 to 2014, pot fishing gear for

sablefish has accounted for 25 and 68% of the fixed gear catch in the AI and BS, respectively (Hanselman *et al.*, 2014). Additionally, the North Pacific Fishery Management Council took final action in April 2015 to legalize pot fishing gear for sablefish in the GOA, largely due to fishermen's accounts of sperm whale depredation (NPFMC April Newsletter, 2015). This management change in response to depredation will likely become effective in the GOA in 2017, yet it remains to be seen what proportion of the fleet will change gear types at that time.

In order to estimate sablefish mortality due to whale depredation, a number of assumptions must be made about historic sablefish catches and the "true" number of sets depredated. Despite these data limitations, there is value in these first estimations of sablefish removals due to whales and for understanding the effects on the sablefish assessment. Overall, the effect of the unaccounted for depredation mortality on the overall estimate of stock status and estimated catch limits was low. However, regionally the unaccounted for mortality can be a substantial portion of the quota. If the stock assessment model is to include these estimates of additional mortality in the fishery, the depredation effect on the survey must also be accounted for. A future management challenge would occur during the apportionment of an increased catch limit among IFQ regions due to accounting for whale depredation on the survey. Whether at the assessment level or at the catch management level, some average level of whale depredation mortality would likely need to be accounted for in those regions. Future research should focus on the estimation of the abundance of depredating whales in relation to changes in the estimates of sablefish stock abundance.

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